



STRATHMORE DATA COMMUNITY

DataSprint 2026 — Participant Brief

What You Need to Know to Get Started

Welcome

You are about to spend one week working with real data about real Kenyans. The dataset you will use comes from the 2024 FinAccess Household Survey — the most comprehensive survey of financial inclusion in Kenya, published in December 2024 by the Central Bank of Kenya, KNBS, and FSD Kenya.

Your task is to build a machine learning model that predicts whether a Kenyan adult's financial situation has Improved, Stayed the same, or Worsened, and to explain what drives the difference. This is a multiclass classification problem.

Dataset Access

Source	Link
Kaggle (recommended — run notebooks directly in browser)	https://www.kaggle.com/datasets/davidpbriggs/kenya-finaccess-household-survey-2024
Official FinAccess site (full report + data manual)	https://finaccess.knbs.or.ke/reports-and-datasets

Download the curated file: finaccess2024_datasprint.csv (20,871 rows × 28 columns)

The Data — Quick Reference

Category	Columns Included
Demographics	county, location_type, Sex, Age, household_size, education_level, marital_status, has_disability
Livelihood	monthly_income, experienced_shock
Mobile & digital	mobile_ownership_1, mobile_money_access, barriers_mobile_money

Category	Columns Included
Financial behaviour	Savings_formal, Savings_informal, Loan_formal, Loan_informal, defaulted, formal_service_use, barriers_bank, prodsum1
Financial health	fl_score, nfhi_11, nfhi_12, nfhi_13, accessto_13k_1month, not_difficult
TARGET	financial_status → Worsened (52.6%) Stayed the same (26.9%) Improved (20.5%)

Missing Values

Column	Missing	What to Do
barriers_bank	27.5%	These are people who have a bank — no barrier. Fill missing values with 'No barrier' before encoding.
monthly_income	~0%	Already imputed with median in the curated file. No action needed.

Suggested Workflow

Step	Task	Tools / Tips
1	Load and inspect the data	pd.read_csv(), df.shape, df.dtypes, df.isnull().sum()
2	Explore — EDA	df.value_counts(), sns.countplot(), sns.heatmap() for correlations
3	Clean and preprocess	Fill missing values, LabelEncoder or pd.get_dummies() for categoricals
4	Split into train and test	train_test_split(df, test_size=0.2, random_state=42)
5	Train a model	Start with LogisticRegression or DecisionTreeClassifier
6	Evaluate	f1_score(y_test, y_pred, average='weighted') — not just accuracy
7	Interpret	feature_importances_, coefficients, or SHAP values
8	Build your slides	7 slides, tell the story, keep it visual

Starter Code

Getting started in Python

```
import pandas as pd
```

```

from sklearn.model_selection import train_test_split
from sklearn.metrics import f1_score, classification_report

df = pd.read_csv('finaccess2024_datasprint.csv')
print(df.shape)  # (20871, 28)
print(df['financial_status'].value_counts())

# Always evaluate with weighted F1 - not accuracy!
score = f1_score(y_test, y_pred, average='weighted')
print(f'Weighted F1: {score:.3f}')

```

Required Deliverables

Item	Required	Notes
Jupyter Notebook (.ipynb)	Yes	Clean, commented, reproducible. Includes full pipeline from loading to evaluation.
Data Visualisations	Yes	Minimum 5 labelled charts in the notebook. Tell a visual story about Kenya.
7-Slide PowerPoint (.pptx)	Yes	Follow the required slide structure below. Keep slides visual — not text-heavy.

Required Slide Structure

Slide	Title	What to Include
1	Problem Statement	What problem are you solving? Why does it matter for Kenya?
2	Dataset Overview	Key statistics, feature categories, target distribution chart
3	Key EDA Finding	Your most interesting insight from data exploration
4	Modelling Approach	Which model? Why? How did you handle preprocessing and class imbalance?
5	Model Performance	Weighted F1-score, classification report, confusion matrix
6	Key Drivers	Which features predict financial status most strongly?
7	Recommendations	What should policymakers, banks, or NGOs do based on your findings?

Tips From iLab Africa

- Start with a simple model first — understand it before trying something complex.
- Visualise before you model — charts will tell you more about the data than any algorithm.

- Think about what the features mean in real life. A monthly_income of KES 1,000 is very different from KES 50,000.
- Class imbalance is real — over half the respondents said ‘Worsened’. Make sure your model learns all three classes.
- The guiding question matters as much as your F1 score — judges will ask you what your model tells policymakers. Know your answer.
- barriers_bank missing values are not random — they represent people who already have bank accounts. Think about how to encode this meaningfully.

Key Dates

Date	Milestone
Thursday, 11th June 2026 4:00 PM	Registration closes
Friday, 12th June 2026 2:00 – 4:30 PM	Kickoff & Info Session MSB 1, Strathmore University
Tuesday, 16th June 2026 12:00 Midday	Submission deadline
Wednesday – Friday, 17th – 19th June 2026	Review period — iLab Africa reviews submissions and provides feedback
Friday, 19th June 2026 2:00 – 5:00 PM	Workshop, Final Presentations, Awards & Networking MSB 1, Strathmore University

Contact

Philip Tait | Chairperson, Strathmore Data Community

datacommunity@strathmore.edu | 0716 914 156 | linktr.ee/strathmoredatacommunity

Learn. Build. Launch.

The Strathmore Data Community is an officially recognised student community at Strathmore University, Nairobi, supported by SCES and SIMS.